Physics - Undergraduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
   (a) 0.1 GeV/c^2
   (b) 0.2 GeV/c^2
   (c) 0.5 GeV/c^2
   (d) 1.0 GeV/c^2

2. The mean kinetic energy of the conduction electrons in metals is ordinarily much higher than kT because
   (a) electrons have many more degrees of freedom than atoms do
   (b) the electrons and the lattice are not in thermal equilibrium
   (c) the electrons form a degenerate Fermi gas
   (d) electrons in metals are highly relativistic

3. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
   (a) 1.2 x 10^9 m/s
   (b) 3.0 x 10^8 m/s
   (c) 1.5 x 10^8 m/s
   (d) 1.0 x 10^8 m/s

4. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
   (a) 4
   (b) 5
   (c) 6
   (d) 20

5. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R. The time required for one revolution is
   (a) independent of M
   (b) proportional to $m^(1/2)$
   (c) linear in R
   (d) proportional to $R^(3/2)$

True / False (10 marks)

Answer True or False

6. True or False: During a reversible thermodynamic process, the temperature of the system must remain constant.
7. True or False: An observer moving at 0.50 c will measure the length of a 1.00 m rod to be 0.80 m due to relativistic effects.
8. True or False: The Sun's energy is primarily produced by the fusion of two hydrogen atoms into one helium atom, resulting in significant energy release.
9. True or False: The electric displacement current through a surface is proportional to the rate of change of the magnetic flux through that surface.
10. True or False: The detector will detect exactly 10 photons when 100 photons are sent, due to its quantum efficiency of 0.1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of the universe's temperature change from 12 K to 3 K on the spatial distribution of galaxies, focusing on their relative distances during these periods.
12. Discuss how the Hall coefficient can be used to determine the sign of charge carriers in a doped semiconductor, including the underlying principles and implications for semiconductor behavior.

End of Paper